

Data Science

Survival Skills

Exercise 2

Exercises + Homeworks



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What you can expect

- Live coding in Python (together with you) to learn DS Survival Skills
- We answer your questions about the homework
- If we still have time, you can do your homework in the exercise session where we can help you directly

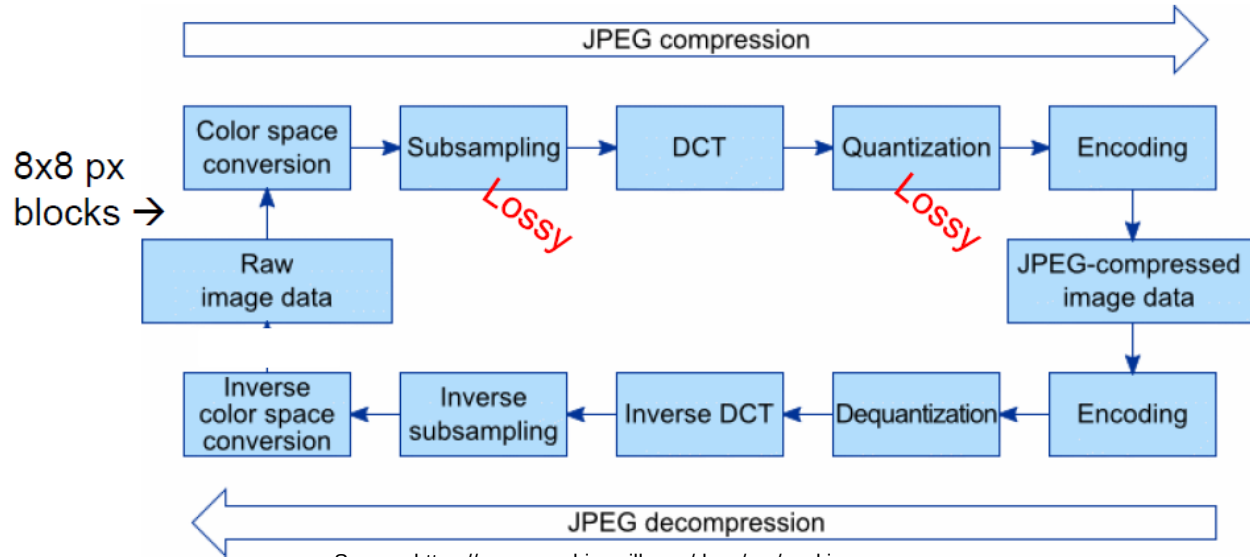
Questions: Getting Started

This exercise: File Handling

- How to load/save files
- File types:
 - Images: .png, .tif, .jpg
 - Video: .mp4
 - JSON
 - .csv
 - .hdf5

Live coding

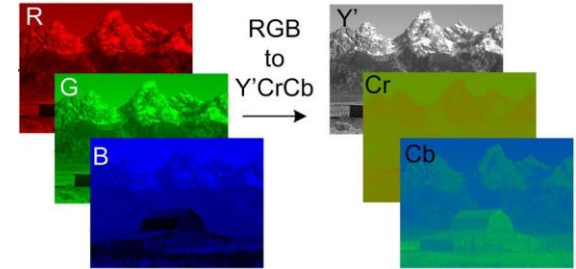
JPEG compression



Source: <https://www.graphicsmill.com/docs/gm/working-with-jpeg.htm>

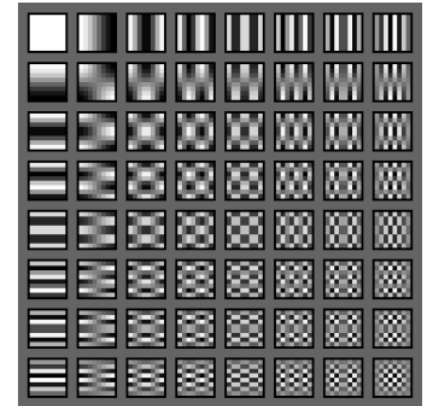
Chroma subsampling: luminance more important than color differences

Color Space Conversion:



Source: https://www.researchgate.net/figure/RGB-to-YCrCb-Conversion_fig2_304625497

Discrete Cosine Transform:



Source: <http://www-i6.informatik.rwth-aachen.de/web/Misc/Coding/365/li/material/notes/Chap4/Chap4.2/Chap4.2.html>

Questions?

- Today's exercise**
- Homework 1 and 2**

And we are done!

Thank you